C4-Helix magic preparation

circuit, state, and fault tolerance (V2)

This note follows one complete preparation attempt in execution order. At each stage it gives the ideal state, the physical circuit, and a fault example explaining the required protection. The protocol uses one gauging/ungauging cycle, three rounds of checks while gauged, and postselection: a failed required test rejects the entire attempt.

This expanded edition keeps the circuit order and examples of V1. Before the first circuit, Sections 2 and 3 state the four proof steps and prove the code lemma used later. Each circuit section then identifies which proof step it establishes. The gauge readout and SWAP correction are worked out together in Sections 12 and 13.

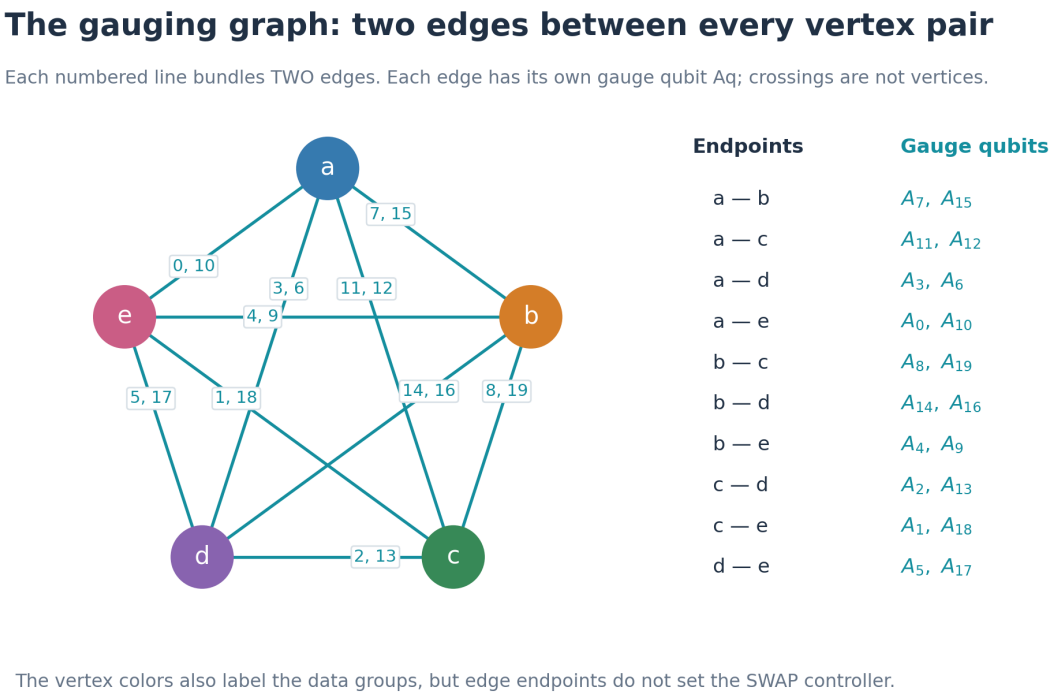


Figure 1: The graph used throughout: each labelled line represents two edges, with one gauge qubit per edge.

The compiled PDF and editable LaTeX contain the same text and vector quantum circuits. This Markdown version embeds images of those circuits.

1 Registers, target state, and the fault allowance

There are twenty data qubits $B_0, \ldots, B_{19}$, another twenty $C_0, \ldots, C_{19}$, twenty gauge ancillas $A_0, \ldots, A_{19}$, and five temporary vertex qubits a,b,c,d,e. Cat leaves are labelled ℓ_j; w is a resettable parity-measurement helper. A bar denotes a logical operator or encoded state.

Each data copy is a $[[20, 2, 6]]$ code. Its inner four-qubit C4 blocks have labels a:0–3, b:4–7, c:8–11, d:12–15, e:16–19. A **matched pair** means the corresponding inner blocks in B and C. For example, matched pair a contains $B_0, \ldots, B_3, C_0, \ldots, C_3$. Two affected matched pairs can therefore contain four actual C4 blocks.

For either data copy D, the original checks are

$$S^X_{D,s} = \prod_{q\in R_s} X_{D_q}, \qquad S^Z_{D,s} = \prod_{q\in R_s} Z_{D_q}, \qquad s = 0,\dots,8.$$

The complete supports R_s are in Appendix A. For example, $R_0 = \{0,1,2,3\}$ gives the inner XXXX and ZZZZ checks. Nine independent checks of each type specify the original code space on each copy. An encoded state is a common $+1$ eigenstate of these checks. Distance six means that a Pauli commuting with every check but changing the encoded information must act on at least six physical qubits. It guarantees correction of two physical-qubit errors, not two arbitrary faults in a circuit.

Prepare the inputs and define the partial and complete SWAPs by

$$|\psi\rangle = |\overline{00}\rangle_B|\overline{+0}\rangle_C,$$
$$T_v = \prod_{q\in J_v} \mathrm{SWAP}(B_q, C_q), \qquad U = T_aT_bT_cT_dT_e, \qquad \Pi_+ = \frac{I+U}{2}.$$

Here J_v is the four-position inner block labelled v. The ideal state after gauging and ungauging is proportional to $|\psi_+\rangle := \Pi_+|\psi\rangle$. The final decoded B readout selects logical $B_0 = 0$, leaving C in

$$|\overline{M0}\rangle_C, \qquad |M\rangle = \frac{2|0\rangle + |1\rangle}{\sqrt{5}}.$$

Fault model and claim. At most two locations fail in the entire attempt. A location is a preparation, gate, physical measurement, or single-qubit waiting interval. A faulty gate may apply an arbitrary error on its participating qubits; reports may be wrong. Classical processing is reliable. Fredkins act on their specified three qubits, or are decomposed on those same qubits. All operations are scheduled serially; unused live qubits have counted idles. Helpers are prepared inside their own gadget and reset before reuse.

The claim is that every accepted attempt within this bound has the correct **decoded** logical output. Correctable physical errors may remain. There is no new two-fault allowance at each stage.

A clean round includes all its helper preparation and idles; boundary idles belong to the following interval. Of any three complete rounds, at least one is clean. This is a fact used in the proof, not information the decoder receives.

2 The four proof steps, before following the circuit

The circuit sections below are in execution order. The four **proof steps** describe what must be established across those sections; they are not four additional operations on the computer.

Proof step 1 - bound how physical damage spreads. Show that each fault can add errors in at most one new matched pair. A matched pair contains one four-qubit inner C4 block in B and its corresponding four-qubit block in C. For example, every SWAP controlled by a has both data targets in $B_0,\dots,B_3,C_0,\dots,C_3$. Damage can move between those two inner blocks, but that controller cannot carry it into b,c,d,e. Verified source preparation and one cat leaf per target port supply the same bound for the other gadgets. Two faults consequently permit at most two matched pairs, or four actual inner C4 blocks. The bound holds term by term in an operator expansion; it does not assume that a general quantum fault is a randomly chosen Pauli.

This step also bounds residual gauge bit errors. It does not certify the logical SWAP outcome: a phase fault can change the operation from $I+U$ to $I-U$ without damaging many physical targets.

Proof step 2 - enforce the intended positive SWAP projection. Prove that the paired, cycle, and Gauss checks in one clean gauged round remove the negative SWAP contribution. The paired checks first reject local errors that could disguise it; the Gauss checks then enforce $U = +1$. Three complete rounds guarantee at least one clean round. After subsequent ungauging the state is a sum of terms

$$E_\nu|\psi_+\rangle, \qquad |\psi_+\rangle = \frac{I+U}{2}|\psi\rangle,$$

where the E_ν obey proof step 1. The data are not yet necessarily free of errors.

Proof step 3 - remove the inherited larger errors. Decode all twenty gauge reports before choosing the ordinary SWAP correction. That restores the original B/C code frame without sending a readout fault into a wrong group-wide correction. Then measure the original checks separately on B and C, in three positive rounds. The two-group lemma proved next shows that a clean round either rejects the inherited errors or leaves the intended encoded state. Errors remaining after the last clean round are due to later faults, with at most one damaged physical data qubit per fault.

Proof step 4 - protect recovery and readout themselves. Use repeated syndrome records to choose a correction despite wrong reports. Decode the complete B-Z readout before evaluating its logical parity. These operations can still fail or introduce new errors, so their faulty locations and C waiting errors remain in the same global budget. The final guarantee concerns the decoded C state; physical correctable errors may remain.

The same circuit fact may serve two proof steps. For instance, full gauge decoding both permits removal of the ancillas in proof step 3 and prevents an incorrect correction from violating the spatial bound in proof step 1.

3 The two-group lemma, with proof

3.1 Statement and the three code facts used

Lemma. On one twenty-qubit C4-Helix copy, a Pauli error supported on at most two inner C4 blocks either anticommutes with an original check, and is detected, or acts as the identity up to phase on every encoded state. It cannot pass all checks while changing the encoded information.

For an arbitrary error operator supported on those same blocks, the component retained by all positive original-code checks is a scalar times the original encoded state. This is a detectability statement: it does not assert that an unknown error on two whole blocks can be corrected without rejection.

The proof uses three properties of this code:

- Each inner block has the checks XXXX and ZZZZ.
- Every full-code generator consists entirely of X or entirely of Z. This is the meaning of CSS here.
- Distance six means that any undetected Pauli acting nontrivially on the encoded information has weight at least six. Weight counts the physical positions on which a Pauli is not I.

The construction and parameters are given in Berthusen and Durso-Sabina, Section 2.1 and Table 1 and in the experimental manuscript's Appendix A. The lemma is derived here from these properties, rather than quoted as a separately named result.

For a valid encoded state $|\varphi\rangle$, every stabilizer S satisfies $S|\varphi\rangle = |\varphi\rangle$. Consequently,

$$(ES)|\varphi\rangle = E|\varphi\rangle.$$

Multiplying E by S is a mathematical replacement with the same action on the code. No physical correction is being performed in this step.

3.2 X-only errors: reduce each block to at most two factors

Assume E contains only X operators and passes all checks. Passing the inner ZZZZ check requires an even number of X factors in each inner block: each X contributes one minus sign.

- Zero X factors: leave that block unchanged.
- Two X factors: leave that block unchanged.
- Four X factors: multiply by its XXXX stabilizer to remove all four, using $(XXXX)(XXXX) = IIII$.

After doing this on both affected blocks, the equivalent operator E' still passes every check, acts identically on encoded states, and has

$$\text{weight}(E') \leq 2 + 2 = 4 < 6.$$

Distance six therefore excludes a nontrivial logical action. The original E also acts trivially. An error failing a check is detected instead.

3.3 Z-only and mixed Pauli errors

For a Z-only error, exchange X and Z in the preceding argument: the inner XXXX check requires an even number of Z factors, and multiplication by ZZZZ removes four. The equivalent error again has weight at most four and cannot be an undetected logical operator.

For a mixed Pauli use $Y = iXZ$ and write

$$E = \omega E_X E_Z,$$

where E_X contains only X operators, E_Z only Z operators, and ω is an overall phase. Both parts occupy the same two allowed inner blocks. If E passes every check, each part does too: against a Z-type check only E_X contributes a commutation sign, and against an X-type check only E_Z does. Their own-type checks commute automatically.

The preceding reductions apply separately to both parts. Each acts trivially on encoded states, so their product does as well. This proves the Pauli statement. Such a Pauli is proportional to a product of stabilizer generators, including the identity.

3.4 Arbitrary errors and application to the two copies

Expand an arbitrary operator on the two inner blocks in Pauli products. Positive measurements project out each term anticommuting with a check. Every remaining term acts as a scalar on the encoded state. Adding those surviving terms gives $c|\varphi\rangle$: reject if $c = 0$; otherwise normalization restores $|\varphi\rangle$.

For the two-copy circuit, two matched pairs permit two inner blocks in B and two in C. Expand the joint error in products of B and C Paulis and apply the lemma to each copy. The conclusion holds even when the encoded B/C state is entangled: the surviving operator acts as a scalar throughout their product code space.

Commuting with checks alone does not bound physical weight. Distance six detects damaging errors below weight six and corrects unknown errors of weight at most two; those are different guarantees. The lemma additionally excludes undetected logical errors on two inner blocks even though those blocks contain eight physical qubits.

3.5 Concrete examples of the distinction

On one copy, the two-block error $F = X_2X_3X_8X_{11}$ is detected by

$$S_6^Z = Z_4Z_7Z_8Z_9Z_{14}Z_{15}Z_{16}Z_{19}, \qquad S_6^Z F = -F S_6^Z,$$

because the overlap is only at position 8. In contrast, $(X_0X_1X_2X_3)(X_8X_9X_{10}X_{11})$ passes the checks but is a product of inner stabilizers and does nothing to the encoded state.

Adding a third inner block changes what is possible:

$$X_2X_3X_8X_{11}X_{12}X_{15} = \overline{X}_0.$$

This logical X passes every check and can change encoded information. The restriction to two inner blocks in the lemma is essential.

4 The measurement circuit used for every check

Ideal preparation. To measure $Q = Q_0 \cdots Q_{m-1}$, with disjoint target ports, prepare one m-leaf cat. A port is one physical qubit for a Pauli factor or one data pair for a SWAP factor. The four-leaf example is

$$|0^4\rangle \xrightarrow{H_{\ell_0},\ \mathrm{CNOT}_{0\to1},\ \mathrm{CNOT}_{1\to2},\ \mathrm{CNOT}_{2\to3}} \frac{|0000\rangle + |1111\rangle}{\sqrt{2}}.$$

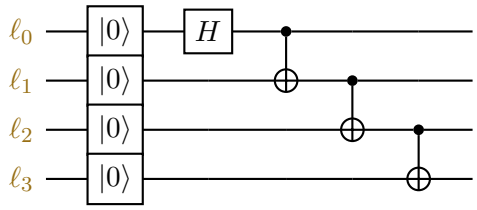


Figure 2: Raw cat preparation. A box containing zero denotes reset.

Extend the chain for larger m. Verify every adjacent ZZ parity three times in complete scans, using the following circuit for each pair. Reset w before each test and require every raw Z report to be zero. Reported bit zero means +1; bit one means -1.

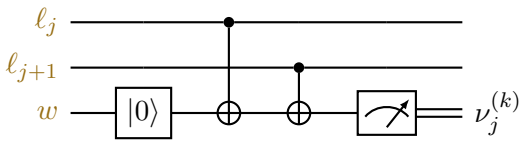


Figure 3: One adjacent-ZZ test. The parity helper w is a CNOT target.

Then leaf j controls Q_j: CNOT for X, CZ for Z, Fredkin for SWAP. After all controlled factors, measure each leaf in X. The circuits show X measurement as H followed by a Z meter. The check bit is

$$y = \lambda_0 \oplus \cdots \oplus \lambda_{m-1}.$$

A positive check requires $y = 0$, not every $\lambda_j = 0$. Each check uses a fresh cat. Each leaf has only one target port. Turquoise labels are gauge qubits; gold labels are cat leaves. A dot and circled plus denote CNOT, two dots denote CZ, and a quantum control with two crosses denotes Fredkin. Double wires

carry classical bits. Panels sharing a cat are parts of one measurement; complete all its controlled factors before reading any of its meters.

Example of the conditional operation. With a possibly imperfect cat $c_0|0^m\rangle + c_1|1^m\rangle$, coupling gives

$$c_0|0^m\rangle|\phi\rangle + c_1|1^m\rangle Q|\phi\rangle.$$

An actual leaf-X sign product $r = \pm 1$ leaves a data state proportional to $(c_0 I + r c_1 Q)|\phi\rangle$. The ideal cat and positive outcome therefore give the positive projector $(I + Q)/2$, up to a common factor for the individual leaf reports.

Proof step 1 - each faulty leaf has one target port. Verification must first prevent errors from raw cat preparation reaching many ports. The calculation below establishes that local bound. The remaining uncertainty in the measured projector is kept for proof steps 2–4; it is not being assumed away.

Fault example and bound. Put X on ℓ_1 after the first cat CNOT but before the second. After the two remaining CNOTs the leaves are in

$$\frac{|0111\rangle + |1000\rangle}{\sqrt{2}}.$$

The error spread to later leaves, but $Z_{\ell_0} Z_{\ell_1} = -1$, so a clean verification rejects it. In general a clean ZZ scan rejects all nonuniform bit patterns and projects onto $\mathrm{span}\{|0^m\rangle, |1^m\rangle\}$. It does not certify the relative phase: the minus cat also passes every ZZ test.

After the clean scan there are no leaf-to-leaf gates. A faulty leaf–w interaction can flip only its one leaf. X on w does not spread back to the controls; Z on w can spread phases. Thus each subsequent fault adds at most one bad leaf bit, and hence one faulty target factor. For example, X on a CNOT control leaf can produce X on its own target, but no other data target.

An operator expansion of a faulty gadget has terms

$$D(aI + bQ).$$

D is the local target damage: at most one target port per fault, also counting separately faulty data idles. The coefficients a,b include cat-phase faults and wrong reports. They need not give the desired projector. For instance, a minus cat with actual positive leaf parity implements $I - Q$ instead of $I + Q$. Repetition of complete later verification rounds must protect against this error; ZZ verification alone does not.

5 Prepare the encoded B and C inputs

The postselected preparation below is illustrative and simplifies the proof; an efficient implementation would instead prepare the known encoded inputs using measurement-dependent Pauli corrections or an encoding circuit, followed by fault-tolerant verification.

Ideal operation. Reset all forty data qubits to zero. For each copy, measure its eighteen original checks and two logical checks, requiring positive outcomes. Execute three complete rounds on B, then three on C:

$$\mathcal{F}_B = (S^X_{B,0:8}, S^Z_{B,0:8}, \overline{Z}^B_0, \overline{Z}^B_1),$$
$$\mathcal{F}_C = (S^X_{C,0:8}, S^Z_{C,0:8}, \overline{X}^C_0, \overline{Z}^C_1).$$

Here 0:8 means the nine checks in increasing s order. The logical supports are listed in Appendix A. The positive projector for a full B source round is $|\overline{00}\rangle\langle\overline{00}|$; for C it is $|\overline{+0}\rangle\langle\overline{+0}|$.

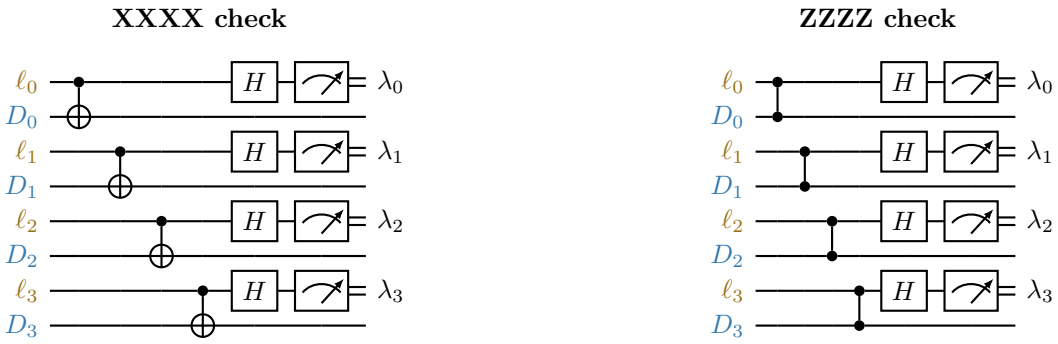


Figure 4: Physical circuits for inner XXXX and ZZZZ checks; D denotes either data copy.

Replace the four targets by R_s for another code check, or by a logical support for an input logical check. Use one shared cat per panel, with six leaves for a logical check and eight for an upper check.

Proof step 1 - initialize the local-error bound. The claim starts after a clean complete source round. Raw preparation may have caused widespread errors before that point, but its accepted projection has a unique output state. The later gadget faults are counted individually.

Fault example. After a correct source state has been prepared, X_{B_0} makes its inner ZZZZ outcome negative. A later clean positive source round rejects this component. A fault in raw preparation can cause a more complicated state; the clean full round projects onto the unique intended source state or rejects it.

Later measurement faults leave local damage. If F is an existing error, Q a measured source stabilizer, and $|\sigma\rangle$ the intended source,

$$(aI + bQ)F|\sigma\rangle = (aF + bQFQ)|\sigma\rangle, \qquad Q|\sigma\rangle = |\sigma\rangle.$$

Pauli conjugation by Q does not enlarge F's support. A gadget's additional D contributes only its faulty targets. Thus residual source errors have at most one physical source-qubit error per fault after its clean round. B waiting while C and the graph are prepared is included in this count.

6 Prepare and verify the graph resource

Ideal operation. Reset the five vertices and twenty A qubits to zero. Apply H to each vertex. For every edge q with endpoints u_q, v_q, apply CNOT from u_q to A_q, then from v_q to A_q. Use increasing q and alphabetic endpoint order. There are forty CNOTs.

The state is

$$|\Gamma\rangle_{VA} = \frac{1}{\sqrt{32}} \sum_{t\in\{0,1\}^5} |t\rangle_V |p(t)\rangle_A, \qquad p(t)_q = t_{u_q} \oplus t_{v_q}.$$

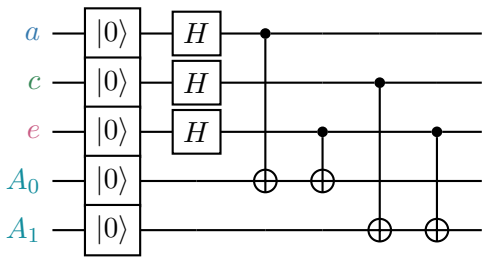


Figure 5: The resource gates for edges 0 (a–e) and 1 (c–e).

The subscript V denotes the five temporary vertices. The twenty-bit string $p(t)$ lists the actual computational-basis values of A. For example, $t = 00000$ and $t = 11111$ both give $p(t) = 0^{20}$: each edge receives either no flips or two flips. Both terms are present in the superposition, together with the other thirty terms.

Verify the 25 checks in the order shown below, in three complete positive rounds. They specify $|\Gamma\rangle$ uniquely:

$$K_v = X_v \prod_{q \in E_v} X_{A_q}, \qquad K_q = Z_{u_q} Z_{A_q} Z_{v_q}, \qquad (K_a, \ldots, K_e, K_0, \ldots, K_{19}).$$

Here E_v is the eight-edge star incident on v. In particular, $E_a = \{0, 3, 6, 7, 10, 11, 12, 15\}$. Each resource-edge check has three ports; each vertex check has nine, including the still-live vertex.

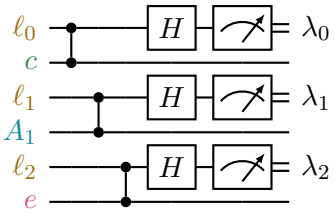


Figure 6: One edge check: K_1 has targets c, A_1, e.

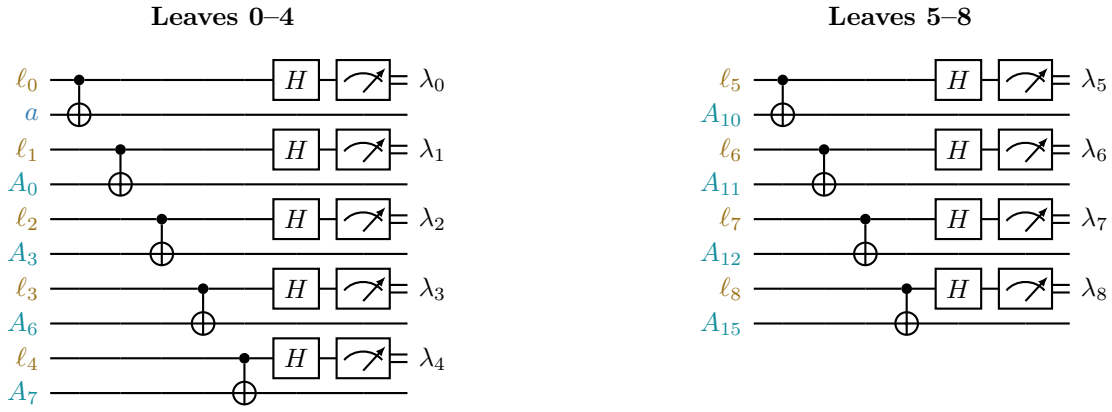


Figure 7: All nine ports of K_a, split across two panels sharing one cat.

Proof step 1 - verify before using a source as a controller. This is necessary both for the data matched-pair bound and for limiting bad A bits before the decoder in proof step 3. The raw graph circuit alone does not have the required one-bit-per-fault property.

Fault example. Let an X fault occur on a after its CNOTs to A_0, A_3, A_6, but before its five remaining CNOTs. Propagation through those remaining gates leaves

$$X_a X_{A_7} X_{A_{10}} X_{A_{11}} X_{A_{12}} X_{A_{15}} |\Gamma\rangle = X_{A_0} X_{A_3} X_{A_6} |\Gamma\rangle.$$

The equality uses $K_a|\Gamma\rangle = |\Gamma\rangle$. One raw fault can therefore produce more than one bad gauge bit. Clean verification rejects this component: K_0, K_3, K_6 are negative.

A clean complete source round projects any accepted input onto $|\Gamma\rangle$. Later Pauli-check gadgets preserve local residual support by the same calculation as in Section 5. A remaining vertex error can affect one matched pair during enrichment; an A bit error affects only its one gauge position. Raw graph preparation is not being assumed fault tolerant by itself.

7 Apply the twenty enrichment Fredkins

Ideal operation. Vertex v controls the four SWAPs with physical labels in J_v. With $V_t = \prod_v T_v^{t_v}$,

$$|\Gamma\rangle|\psi\rangle \longmapsto \frac{1}{\sqrt{32}} \sum_t |t\rangle_V |p(t)\rangle_A V_t |\psi\rangle.$$

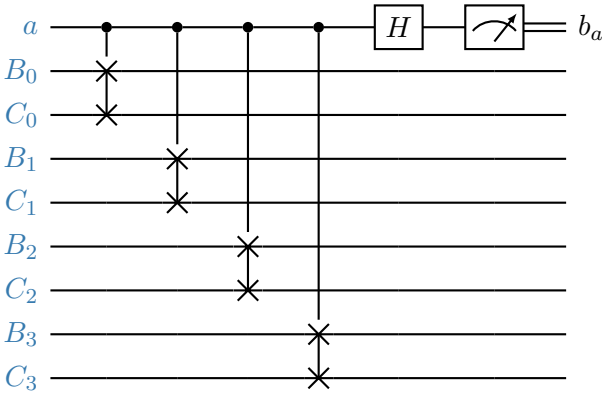


Figure 8: Four Fredkins controlled by a. Complete all twenty Fredkins before any vertex meter is read.

The controller assignment is a:0–3, b:4–7, c:8–11, d:12–15, e:16–19. It differs from the edge endpoints. For example, A_1 has endpoints c,e, while the pair B_1, C_1 is controlled by a.

Example of the superposition. The vertex strings $t = 01000$ and $\bar{t} = 10111$ give the same A string: ones on the eight edges touching b, zeros elsewhere. Their data states after enrichment are, respectively,

$$T_b|\psi\rangle, \qquad T_a T_c T_d T_e |\psi\rangle = T_b U |\psi\rangle.$$

These are two components passing through one circuit. The bar flips all five vertex bits; it does not denote logical encoding. All physical SWAPs here act on disjoint data pairs and square to I. Multiplying V_t by U therefore removes the SWAPs for which $t_v = 1$ and adds those for which $t_v = 0$: the result is $V_{\bar{t}}$.

Proof step 1 - SWAPs preserve the matched-pair label. Each Fredkin controlled by v has data targets B_q, C_q with $q \in J_v$. These are corresponding positions in the two four-qubit C4 blocks labelled v. A fault may spread to several of those positions, but none of that controller's data couplings reaches another label.

Fault example. An X fault on a after its first Fredkin is propagated through the remaining three as an X on a times ordinary SWAPs on pairs 1,2,3. For one controlled-SWAP gate F,

$$F X_a F = X_a \ \mathrm{SWAP}(B_q, C_q).$$

Several data pairs can therefore be damaged, but all are in matched pair a. A faulty Fredkin can also damage its own data targets; they belong to the same matched pair as its controller. Existing data errors may exchange B/C positions, but not matched-pair labels. This is why one controller is assigned to only one matched pair.

8 Read the vertices and apply the ancilla-Z corrections

8.1 These operations are required even without faults

Measuring the vertices and selecting the positive SWAP outcome are part of the **noiseless protocol**. The known measurement-dependent phases must also be accounted for even with perfect preparations, gates, and measurements. This note applies physical Z gates so that every later Gauss check is expected to be positive. Alternatively, those known phases can be tracked classically if the expected check signs are adjusted, as explained in Section 8.5.

The Z gates remove known signs introduced by the vertex measurements. This is **feedforward**: a recorded measurement result determines the next gate. It is different from recovering an unknown error using a syndrome. Sections 9–11 add checks against faults; Section 8.7 explains what those checks must protect at this stage.

The vertices are measured in **X**. In the circuit, this means applying H and then measuring Z on each vertex. Thus a Z meter following H implements the X measurement discussed here. The edge ancillas $A_0, \ldots, A_{19}$ are not yet measured in Z; that readout occurs in Section 12.

8.2 The state immediately after reading the vertices

Recall the ideal state at the end of enrichment:

$$\frac{1}{\sqrt{32}} \sum_{t\in\{0,1\}^5} |t\rangle_V |p(t)\rangle_A V_t |\psi\rangle.$$

Here $t = (t_a, t_b, t_c, t_d, t_e)$ contains the five vertex bits. An edge joining vertices v and w contains the bit $p(t)_q = t_v \oplus t_w$; the twenty bits together form $p(t)$. The operator $V_t = \prod_v T_v^{t_v}$ swaps the matched four-qubit blocks for which $t_v = 1$. For example, $V_{01010} = T_b T_d$. The full data SWAP is $U = T_a T_b T_c T_d T_e$.

Measure each vertex in X and record

$$b_v = 0 \text{ for outcome } +1, \qquad b_v = 1 \text{ for outcome } -1.$$

The overlaps with a computational-basis bit t_v are

$$\langle +|t_v\rangle = \frac{1}{\sqrt{2}}, \qquad \langle -|t_v\rangle = \frac{(-1)^{t_v}}{\sqrt{2}}.$$

Consequently, for a fixed five-bit report $b = (b_a, b_b, b_c, b_d, b_e)$, the remaining qubits have the conditional state

$$|\chi_b\rangle = \frac{1}{32} \sum_t (-1)^{b_a t_a + b_b t_b + b_c t_c + b_d t_d + b_e t_e} |p(t)\rangle_A V_t |\psi\rangle.$$

This state is unnormalized: its squared norm is the probability of that report. Even a perfect measurement introduces the displayed outcome-dependent signs.

8.3 Even parity selects the positive SWAP outcome

Compute the bit

$$r = b_a \oplus b_b \oplus b_c \oplus b_d \oplus b_e.$$

Require even parity means keep the attempt only if $r = 0$: zero, two, or four vertices gave outcome -1. It does not mean that all five outcomes must be $+1$. The bit r takes values 0 or 1; the corresponding measured SWAP eigenvalue is $(-1)^r$, taking values $+1$ or -1.

To see why this parity measures SWAP, pair each vertex pattern t with its bitwise complement $\bar{t} = t \oplus (1,1,1,1,1)$. They have

$$p(\bar{t}) = p(t), \qquad V_{\bar{t}} = V_t U.$$

They therefore contribute to the same edge-ancilla ket. Their measurement signs differ by

$$(-1)^{\sum_v b_v(1-t_v)} = (-1)^r (-1)^{\sum_v b_v t_v}.$$

Their combined contribution to $|\chi_b\rangle$ is thus

$$\frac{1}{32}(-1)^{\sum_v b_v t_v} |p(t)\rangle_A V_t (I + (-1)^r U)|\psi\rangle.$$

Hence $r = 0$ selects $I+U$, whereas $r = 1$ selects $I-U$. This is an ideal measurement outcome, not an error syndrome. An odd result can occur without any fault. Section 8.6 explains why this magic-preparation protocol selects the positive outcome.

8.4 Apply the known ancilla-Z correction

Even for $r = 0$, different t terms may still have different signs. Remove them by applying

$$F_b = Z_{A_7}^{b_b} Z_{A_{11}}^{b_c} Z_{A_3}^{b_d} Z_{A_0}^{b_e}.$$

Here $Z^0 = I$ and $Z^1 = Z$. These are physical Z gates controlled by classical reports. The four selected edges join the root a to b,c,d,e, respectively, so their bits are

$$p(t)_7 = t_a \oplus t_b, \quad p(t)_{11} = t_a \oplus t_c, \quad p(t)_3 = t_a \oplus t_d, \quad p(t)_0 = t_a \oplus t_e.$$



Figure 9: The four classical report-to-Z corrections. Double wires carry measured bits.

Example. Let the report be

$$(b_a, b_b, b_c, b_d, b_e) = (0, 1, 0, 1, 0).$$

Its parity is even, so it is accepted. The measurements have multiplied each t term by $(-1)^{t_b + t_d}$. Apply $Z_{A_7} Z_{A_3}$. These gates contribute

$$(-1)^{(t_a \oplus t_b) + (t_a \oplus t_d)} = (-1)^{t_b + t_d},$$

which cancels the measurement sign.

For a general report b, the product of the measurement sign and the correction sign is

$$(-1)^{\sum_v b_v t_v}(-1)^{\sum_{v\neq a} b_v(t_a\oplus t_v)} = (-1)^{r t_a}.$$

All sign exponents are evaluated modulo two. The non-root terms cancel in pairs; the remaining coefficient of t_a is the parity r. For an accepted report, $r = 0$, so every term has sign $+1$. There is no separate correction controlled by b_a: the even-parity condition already fixes it from the other four reports.

Reset the measured vertices for helper reuse.

8.5 The corrected gauged state and its check signs

For any report, combining each t term with its complement gives

$$F_b|\chi_b\rangle = \frac{1}{32}\sum_{t:t_a=0} |p(t)\rangle_A V_t(I + (-1)^r U)|\psi\rangle.$$

Choosing the member with $t_a = 0$ only labels each pair once; it does not measure or reset a vertex, and it does not discard the complementary contribution. For accepted even parity, this becomes

$$\frac{1}{16}\sum_{t:t_a=0} |p(t)\rangle_A V_t|\psi_+\rangle, \qquad |\psi_+\rangle = \frac{I+U}{2}|\psi\rangle.$$

This is the same conditional state for every accepted ideal report. The two computational components supplied $V_t|\psi\rangle$ and $V_t U|\psi\rangle$; X readout combines them, and F_b removes their remaining known signs.

Why the correction matters for the later checks. The Gauss check at vertex v is

$$G_v = \left(\prod_{q:\, q \text{ touches } v} X_{A_q}\right) T_v.$$

Before correction, the ideal conditional state satisfies $G_v|\chi_b\rangle = (-1)^{b_v}|\chi_b\rangle$. This follows from the stabilizer $X_v G_v$ of the enriched state: measuring X_v with outcome $(-1)^{b_v}$ leaves that same sign for G_v. These signs reflect the ideal measurement results, not unknown faults.

A Z on an edge flips the Gauss-check signs at both endpoints of that edge. The correction F_b makes the signs at b,c,d,e all positive; the remaining sign at a is $(-1)^r$, also positive on an accepted attempt. For the example $b = 01010$, Z_{A_7} flips the signs at a and b, and Z_{A_3} flips those at a and d. The sign at a is flipped twice, while the negative signs at b and d are removed.

Without F_b, even parity would still select the positive SWAP component, but some Gauss checks could be negative solely because of the ideal measurement outcomes. The correction puts every accepted ideal outcome into the prescribed state with all Gauss signs positive.

If the known correction were omitted. Consider again the even-parity vertex report

$$(b_a, b_b, b_c, b_d, b_e) = (0, 1, 0, 1, 0).$$

Let $|\Phi_+\rangle$ denote the intended joint state of the gauge ancillas and data, with all Gauss checks positive (the reference vector defined in Section 8.7). Without the correction $Z_{A_7}Z_{A_3}$, the state would instead be proportional to

$$Z_{A_7}Z_{A_3}|\Phi_+\rangle.$$

The first Z flips the Gauss-check signs at a and b; the second flips them at a and d. The two flips at a cancel. Writing g_v for the outcome of measuring G_v, the uncorrected state therefore gives

$$(g_a, g_b, g_c, g_d, g_e) = (+1, -1, +1, -1, +1).$$

Our current protocol would reject this otherwise valid attempt, because it requires every Gauss check to be positive. These negative outcomes would be caused by the known vertex-measurement signs, even though no fault had occurred.

This does **not** change the positive SWAP projection into the negative one. The vertex parity is even, and $Z_{A_7}Z_{A_3}$ acts only on the gauge ancillas, so the state still satisfies $U = +1$.

To see what would happen without the intervening Gauss tests, suppose we proceeded directly through perfect edge-Z readout and ungauging. Let $m_7, m_3 \in \{0, 1\}$ be the recorded edge bits, with zero for Z outcome $+1$ and one for outcome -1. The omitted correction would leave the scalar sign $(-1)^{m_7+m_3}$ in that measurement branch. After reading the edges, the data state would be proportional to $(-1)^{m_7+m_3}V_t|\psi_+\rangle$, and applying the decoded correction would give

$$V_t\big[(-1)^{m_7+m_3}V_t|\psi_+\rangle\big] = (-1)^{m_7+m_3}|\psi_+\rangle.$$

For this fixed measurement record, the sign is an irrelevant global phase. The final projected data state would be unchanged. The purpose of physically applying F_b is therefore to put the intermediate state into our chosen **all-positive Gauss-check convention**, rather than to repair an unknown fault or to change the ideal SWAP outcome.

One could instead record the known phases classically and adjust the expected Gauss-check signs; for this example, the expected signs at b and d would be negative. Simply omitting the Z gates while retaining the all-positive acceptance rule would cause unnecessary rejection. The circuit and fault-tolerance proof in this note use the physical correction F_b.

8.6 Why this protocol selects the positive outcome

Gauging followed by ungauging can in principle retain both outcomes and record which projector was applied:

$$\Pi_+ = \frac{I+U}{2} \quad (r=0), \qquad \Pi_- = \frac{I-U}{2} \quad (r=1).$$

This is a measurement of U. The original gauging manuscript likewise retains the symmetry-measurement eigenvalues in its final projectors (see its discussion of gauging and ungauging near the end of the source manuscript). Selecting $+1$ here is a choice of our magic-preparation protocol, not a requirement for defining the general measurement.

For our chosen input, however, the negative branch supplies no magic. The input is $|\overline{00}\rangle_B|\overline{+0}\rangle_C$. Both second logical qubits remain in $|0\rangle$, so omit them and write the first logical pair simply as $|0+\rangle$. Its normalized negative branch is

$$\frac{(I-U)|0+\rangle}{\|(I-U)|0+\rangle\|} = \frac{|01\rangle - |10\rangle}{\sqrt{2}}.$$

This is a stabilizer Bell state. Measuring B in Z leaves C in $|1\rangle$ or $|0\rangle$, neither a magic state. In contrast, the positive branch gives

$$\frac{(I+U)|0+\rangle}{\|(I+U)|0+\rangle\|} = \frac{2|00\rangle + |01\rangle + |10\rangle}{\sqrt{6}}.$$

Selecting logical B zero then gives $(2|0\rangle + |1\rangle)/\sqrt{5}$ in C, as worked out in Section 16. Thus we select the positive outcome for this particular input and output resource. Keeping both outcomes would also

require changing the later Gauss-check acceptance rules; the present protocol and proof require all those signs to be positive.

8.7 What the fault-tolerance proof must add

The preceding operations assume accurate reports and perfect gates. They remove known measurement-dependent signs. They do not by themselves certify that an unknown fault has not changed the state. A wrong report can select an incorrect F_b, a faulty correction gate can damage an edge ancilla, and a phase fault can change the SWAP projection itself.

For the subsequent proof define the reference vectors

$$|\psi_\pm\rangle = \frac{I \pm U}{2}|\psi\rangle, \qquad |\Phi_\pm\rangle = \frac{1}{4}\sum_{t:t_a=0} |p(t)\rangle_A V_t |\psi_\pm\rangle.$$

They may be unnormalized; the corrected ideal branch above is $|\Phi_+\rangle/4$. U acts only on the data and commutes with every V_t, hence $U|\Phi_\pm\rangle = \pm|\Phi_\pm\rangle$.

Proof steps 1 and 2 - separate physical damage from the measured projection. Local damage is still bounded by matched pairs. Phase faults and wrong reports need a different argument: the spatial bound alone cannot force equal coefficients of I and U.

Fault example. A Z fault on a just before its X measurement multiplies the t component by $(-1)^{t_a}$. For reported b=00000, the two terms t and its complement now combine with a minus sign: the branch can contain $|\Phi_-\rangle$ instead of $|\Phi_+\rangle$. This does not require many damaged data qubits. The measured SWAP projection itself is wrong. The checks in the next sections are needed to detect it; the spatial error bound alone cannot do so.

More general wrong amplitudes can depend on t_b, t_c, t_d, t_e when the representative has $t_a = 0$. Any function of these four bits is a linear combination of the sixteen products of their signs $(-1)^{t_v}$. On A, those signs are supplied by $Z_{A_7}, Z_{A_{11}}, Z_{A_3}, Z_{A_0}$, respectively, since the four tree-edge bits equal t_b, t_c, t_d, t_e. Thus those amplitude errors can be expanded as gauge-Z operators. Including the bounded gauge-bit and local data errors, the state before a clean round is a sum of terms $F_A E|\Phi_\pm\rangle$, where F_A acts only on A. Its X part has weight at most the number of faults; its Z part can be unrestricted. E occupies at most one new matched-pair label per fault. This accounts for unequal or missing contributions rather than assuming they already combine correctly.

9 While gauged: measure the eighteen paired checks

Ideal operation. Measure, in order, the nine X products and nine Z products

$$N_s^P = S_{B,s}^P S_{C,s}^P, \qquad s = 0, \ldots, 8, \quad P = X, Z.$$

Use one cat across both copies for each check, with B targets first. All leaf-X parities must be zero. Both $|\Phi_+\rangle$ and $|\Phi_-\rangle$ satisfy these paired checks: the original inputs do, and matched factors $P_{B_q} P_{C_q}$ commute with SWAP.

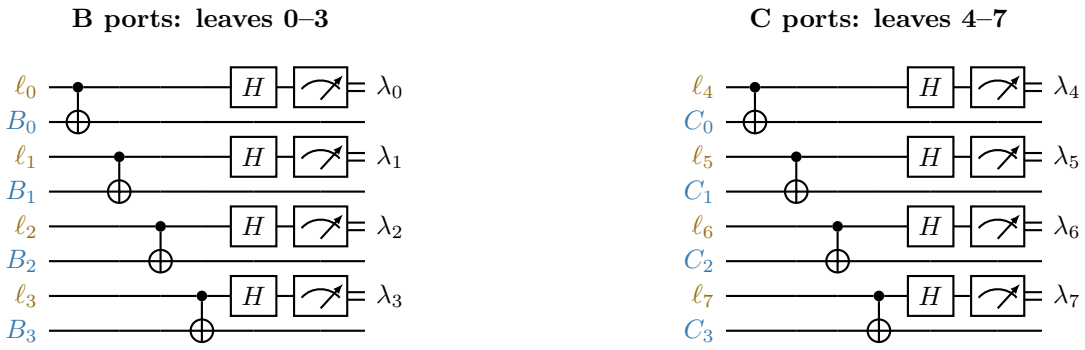


Figure 10: One eight-leaf cat for N_0^X. The panels yield one combined parity.

For an eight-position upper support R_s, the cat has sixteen leaves. Z products use CZ gates instead of CNOTs. Individual B and C stabilizers are not measured separately at this stage.

Error examples. X_{B_0} flips the paired inner-Z check and is rejected by a clean positive measurement. In contrast, keep the following example through the rest of the protocol:

$$E = X_{B_0} X_{C_1} X_{C_2} X_{C_3}.$$

It gives one minus sign on B and three on C, so their paired inner-Z product stays positive. In fact it passes every paired check. Write its one-copy patterns as $P = X_0$, $R = X_1 X_2 X_3$. Then

$$P^\dagger R = X_0 X_1 X_2 X_3 = S, \qquad E = (P_B P_C) S_C.$$

Here S is the inner XXXX stabilizer. P_C is the same pattern as P applied to C; it cancels an extra X_{C_0} in S_C. P on B has not been changed. The dagger is the inverse of P.

Proof step 2, using the lemma from Section 3. The code has the following two-group property: a Pauli on at most two inner C4 blocks of one copy, commuting with all its checks, is a stabilizer up to phase. To see it, separate its X and Z parts, using $Y = iXZ$. Inner checks force an even number of each per block. Multiplying by inner XXXX or ZZZZ removes four equal factors, leaving each part at weight at most four. Since every generator contains only X or only Z, both parts individually commute with all checks. Distance six excludes a nontrivial logical action for either part.

For a general two-matched-pair Pauli $E = P_B R_C$, passing paired checks means P and R have the same commutation sign with every original check. Thus $P^\dagger R$ is an undetected one-copy Pauli on at most two inner blocks, and the property just proved gives $P^\dagger R = S$. Consequently $E = (P_B P_C) S_C$, up to phase. Here the common pair labels from proof step 1 are essential: two unrelated labels in B and two different labels in C would give up to four labels in $P^\dagger R$, outside the lemma's hypothesis.

On states passing paired checks, $S_B S_C = +1$, hence S on B and C act identically. The factor $P_B P_C$ is unchanged by exchanging copies. It follows that

$$UE|\chi\rangle = EU|\chi\rangle \quad \text{if all paired checks on } |\chi\rangle \text{ are positive.}$$

For the worked example this can be checked without the general argument:

$$UEU = X_{C_0} X_{B_1} X_{B_2} X_{B_3} = E N_0^X.$$

On paired-positive states, E and its exchanged version therefore act identically. A surviving error cannot convert a negative U eigenvalue to a positive one. Phase and bit operators on A commute with U and do not change this conclusion.

10 While gauged: measure the sixteen cycle checks

Ideal operation. For each cycle support η, measure $L_\eta = \prod_{q\in\eta} Z_{A_q}$, with one fresh cat and CZ couplings. The ideal strings $p(t)$ give $+1$: along a closed cycle, each vertex bit occurs twice in the XOR of the edge bits.



Figure 11: Parallel-edge and triangle cycle-check circuits.

The ordered supports are

$$(7,15),(11,12),(3,6),(0,10),(8,19),$$
$$(14,16),(4,9),(2,13),(1,18),(5,17),$$
$$(7,8,11),(7,14,3),(7,4,0),$$
$$(11,2,3),(11,1,0),(3,5,0).$$

Proof step 2 - check the ancilla bit constraints. The cycle tests are part of the same clean gauged round as the paired and Gauss tests. Their bit constraints also support the decoding argument in proof step 3.

Fault example. X_{A_7} changes the sign of $L_{(7,15)} = Z_{A_7}Z_{A_{15}}$, so a clean positive cycle check rejects it. A gauge Z error does not change this outcome; cycle checks certify bit consistency, not all possible phase errors.

These sixteen independent constraints characterize the sixteen valid strings $p(t)$. Different such strings are separated by at least eight positions, as proved in Section 12. Thus a nonzero X pattern of weight one or two cannot move one valid string to another; a clean complete cycle measurement rejects those terms. Faulty cycle gadgets touch only individual A ports, while their cat-phase uncertainty is allowed in the coefficients of $aI + bL_\eta$.

11 While gauged: Gauss checks protect the SWAP projection

Ideal operation. For v=a,b,c,d,e, measure

$$G_v = \left(\prod_{q\in E_v} X_{A_q}\right) T_v.$$

Use one twelve-leaf cat: eight CNOT ports to the gauge star, then four Fredkin ports to the corresponding data pairs. Measure all twelve leaves in X only after all controlled factors. Require their one parity to be zero.

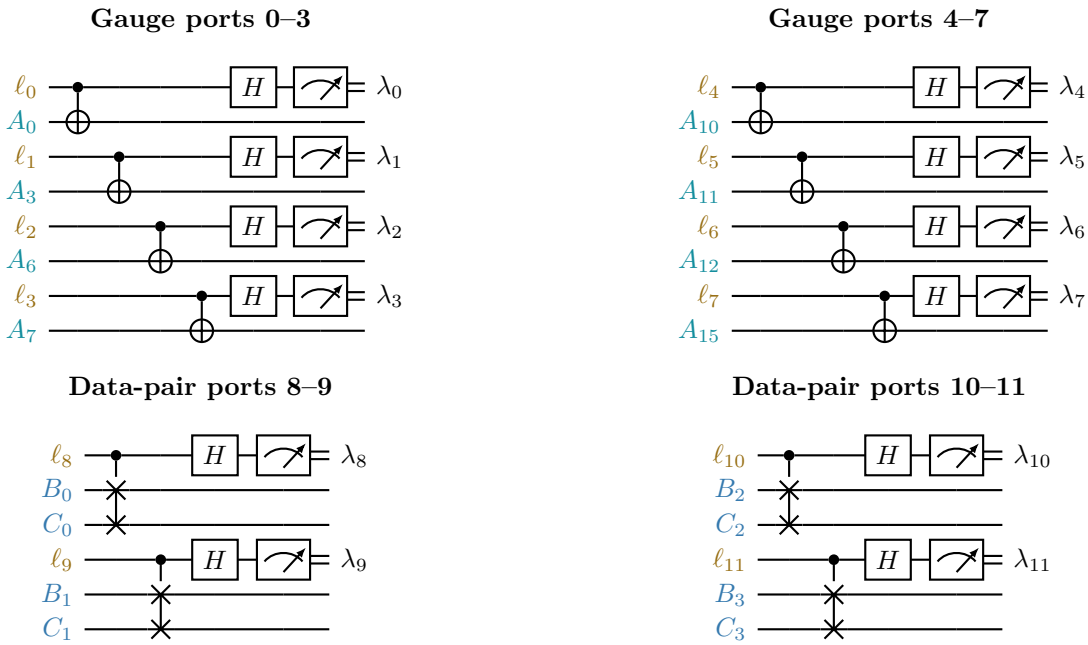


Figure 12: All twelve ports of G_a. All four panels share one cat.

A Gauss operation flips the star bits and applies the corresponding partial data SWAP together. On a component $|p(t)\rangle_A V_t|\phi\rangle$, it toggles the label t_v consistently. Its eight gauge flips are not eight independent errors. Relabelling the sixteen terms gives

$$G_a|\Phi_\pm\rangle = \pm|\Phi_\pm\rangle, \qquad G_v|\Phi_\pm\rangle = |\Phi_\pm\rangle \quad (v = b, c, d, e).$$

For a, toggling the root requires complementing the representative to make the root zero again, adding U to the data; this gives the sign above. For a nonroot vertex it only permutes the sixteen terms.

Proof step 2 - the accepted clean round enforces $U = +1$. The paired checks prevent a bounded local error from disguising a negative SWAP sign. These Gauss measurements now select that sign, while their local couplings continue to obey proof step 1.

Examples of the protection. The vertex-Z fault in Section 8 can leave $|\Phi_-\rangle$. A clean G_a measurement gives -1 and rejects it. The worked data error E of Section 9 still passes on the positive state:

$$G_a E G_a = E N_0^X, \qquad G_a E|\Phi_\pm\rangle = \pm E|\Phi_\pm\rangle.$$

E cannot hide the negative sign; it may remain on a positive state. For comparison, a Gauss measurement alone acting on an X_{B_0} error gives

$$\frac{I + G_a}{2} X_{B_0}|\Phi_+\rangle = \frac{X_{B_0} + X_{C_0}}{2}|\Phi_+\rangle.$$

The physical damage can remain or become a superposition, but is still inside matched pair a. In a clean complete round the paired checks would reject that single-X error first.

Why all three check families are repeated together. A full gauged round consists of 18 paired checks, 16 cycle checks, then five Gauss checks. Execute three such rounds. These 39 observables commute. Their positive Gauss projector contains the positive SWAP projector because

$$G_a G_b G_c G_d G_e = U, \qquad \prod_v \frac{I + G_v}{2} = \left[\prod_{v=b}^{e} \frac{I + G_v}{2}\right] \Pi_+.$$

In a clean accepted round, paired measurements first eliminate Pauli terms that fail their checks. Each surviving E preserves the U sign, by Section 9, so

$$\Pi_+ E|\Phi_-\rangle = 0, \qquad \Pi_+ E|\Phi_+\rangle = E|\Phi_+\rangle.$$

There is at least one clean complete round under the two-fault budget. It removes the negative reference contribution. The remaining projectors preserve the local-error bound. Indeed every measured Q stabilizes $|\Phi_+\rangle$, so clean or faulty subsequent check operations obey

$$D(aI + bQ)F|\Phi_+\rangle = D(aF + bQFQ)|\Phi_+\rangle.$$

Pauli conjugation does not spread support; Gauss conjugation only exchanges B and C within matched pairs. D adds only its own faulty ports. Thus later false reports cannot restore a free negative reference component: their remaining damage stays in the bounded error operators multiplying $|\Phi_+\rangle$. No fresh fault allowance begins after the clean round.

12 Read and decode the twenty gauge bits

Section 12 determines which correcting SWAPs to apply. After measuring the edge ancillas, the ideal data state is

$$V_t|\psi_+\rangle.$$

We decode the twenty measurement reports to reconstruct $t = (0, t_b, t_c, t_d, t_e)$. Each bit equal to one identifies a matched pair of four-qubit blocks to swap.

Section 13 physically applies those SWAPs, namely V_t. Since each SWAP is its own inverse,

$$V_t(V_t|\psi_+\rangle) = |\psi_+\rangle.$$

For example, if Section 12 finds $t = (0, 1, 0, 1, 0)$, the data carry $T_b T_d$. Section 13 applies $T_b T_d$ again, cancelling it.

The additional purpose of decoding in Section 12 is to prevent wrong measurement reports from selecting the wrong correcting SWAPs.

Ideal operation. Measure every A_q in Z, in increasing q order, giving reports $m_0, \ldots, m_{19}$. Decode all twenty before deciding which data SWAPs to apply.

12.1 The reported bits and the bits being estimated

The physical Z measurements give a recorded twenty-bit vector

$$m = (m_0, \ldots, m_{19}), \qquad m_q = 0 \text{ for reported } +1, \quad m_q = 1 \text{ for reported } -1.$$

The letter m always denotes these reports, including possible wrong bits. In an ideal component of the quantum state, the actual ancilla bits are $p(t)_q = t_{u_q} \oplus t_{v_q}$, as prepared in Section 6. The decoder estimates that valid vector; call the estimate $\widehat{p}$. The hat means an estimate made from m, not another physical measurement.

We compare a candidate p with m by counting differing positions:

$$d_H(m, p) = \#\{q \in \{0, \ldots, 19\} : m_q \neq p_q\}.$$

This is the Hamming distance. For example, a single wrong report at position 7 gives distance one from the underlying valid vector.

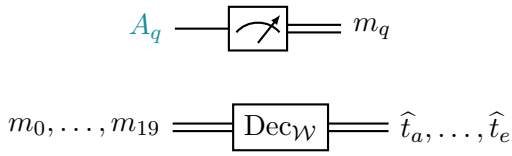


Figure 13: Individual gauge-Z readout, followed by classical full-word decoding.

12.2 Why choose the reconstructed root bit to be zero?

The edge measurements cannot distinguish the vertex pattern t from its complement $\bar{t}$, obtained by flipping all five bits. Both patterns give the same edge bits, because flipping both endpoints leaves their XOR unchanged.

To choose one definite correction, use the pattern whose bit at vertex a is zero. If a candidate pattern starts with zero, keep it. If it starts with one, flip all five bits.

For example, the patterns

$$(t_a, t_b, t_c, t_d, t_e) = (1, 0, 1, 0, 1) \quad \text{and} \quad (0, 1, 0, 1, 0)$$

give identical edge bits. We choose $(0, 1, 0, 1, 0)$, because its first bit is zero. The corresponding correction is $V_t = T_b T_d$.

The decoder therefore tests the sixteen patterns

$$t = (0, t_b, t_c, t_d, t_e), \qquad t_b, t_c, t_d, t_e \in \{0, 1\}.$$

Setting $t_a = 0$ is only a convention for choosing the correction. It does not assert that the original vertex was in state $|0\rangle$, and it does not require another measurement.

The reason this is safe for the data will be shown explicitly in Section 13: the two possible corrections differ by the complete SWAP U, which acts as $+1$ on the intended positive-projection state.

12.3 The actual classical decoding rule

The classical box in the drawing uses the set $\mathcal{W} = \{p(t) : t_a = 0\}$. Enumerate its sixteen strings. Choose the string $\widehat{p}$ differing from m in at most two positions; reject if none exists. From the decoded string set

$$\widehat{t}_a = 0, \quad \widehat{t}_b = \widehat{p}_7, \quad \widehat{t}_c = \widehat{p}_{11}, \quad \widehat{t}_d = \widehat{p}_3, \quad \widehat{t}_e = \widehat{p}_0.$$

These four edges join the root a to the other vertices. The formulas follow directly from their endpoints:

$$p(t)_7 = t_a \oplus t_b, \quad p(t)_{11} = t_a \oplus t_c,$$
$$p(t)_3 = t_a \oplus t_d, \quad p(t)_0 = t_a \oplus t_e.$$

When the representative has $t_a = 0$, each of these edge bits equals the other endpoint's bit. First correct the entire report m to $\widehat{p}$; only then read these four positions to obtain $\widehat{t}$.

Fault example. Suppose the actual A string is 0^{20}, but the report m_7 is wrong and equals one. Reading that tree bit directly would wrongly select T_b. Full-word decoding returns 0^{20}, so the correction contains no SWAPs. An existing X_{B_2} data error is not removed by this classical decoding; it is still handled later.

12.4 Worked example: reconstruct b and d despite a wrong tree bit

Consider the valid representative and its complement

$$t = (0, 1, 0, 1, 0), \qquad \bar{t} = (1, 0, 1, 0, 1).$$

For either choice the ancillas have ones at exactly these positions:

$$\{q : p(t)_q = 1\} = \{2, 3, 4, 5, 6, 7, 8, 9, 13, 15, 17, 19\}.$$

For example, edge 7 joins a and b, so $p_7 = 0 \oplus 1 = 1$. Edge 14 joins b and d, so $p_{14} = 1 \oplus 1 = 0$. In increasing physical order $q = 0, \ldots, 19$, the full vector is

$$p(t) = (0, 0, 1, 1, 1, 1, 1, 1, 1, 1, 0, 0, 0, 1, 0, 1, 0, 1, 0, 1).$$

Suppose the only wrong report is $m_7 = 0$ instead of 1; all other reported bits agree with this vector. Its distance from the valid candidate above is one, so the decoder returns $\widehat{p} = p(t)$. Uniqueness within radius two is proved below. The reconstructed bits are then

$$\widehat{t} = (0, \widehat{p}_7, \widehat{p}_{11}, \widehat{p}_3, \widehat{p}_0) = (0, 1, 0, 1, 0).$$

The correction therefore applies $T_b T_d$. Reading the noisy report m_7 directly would instead give the wrong b bit and apply only T_d, omitting all four required b SWAPs.

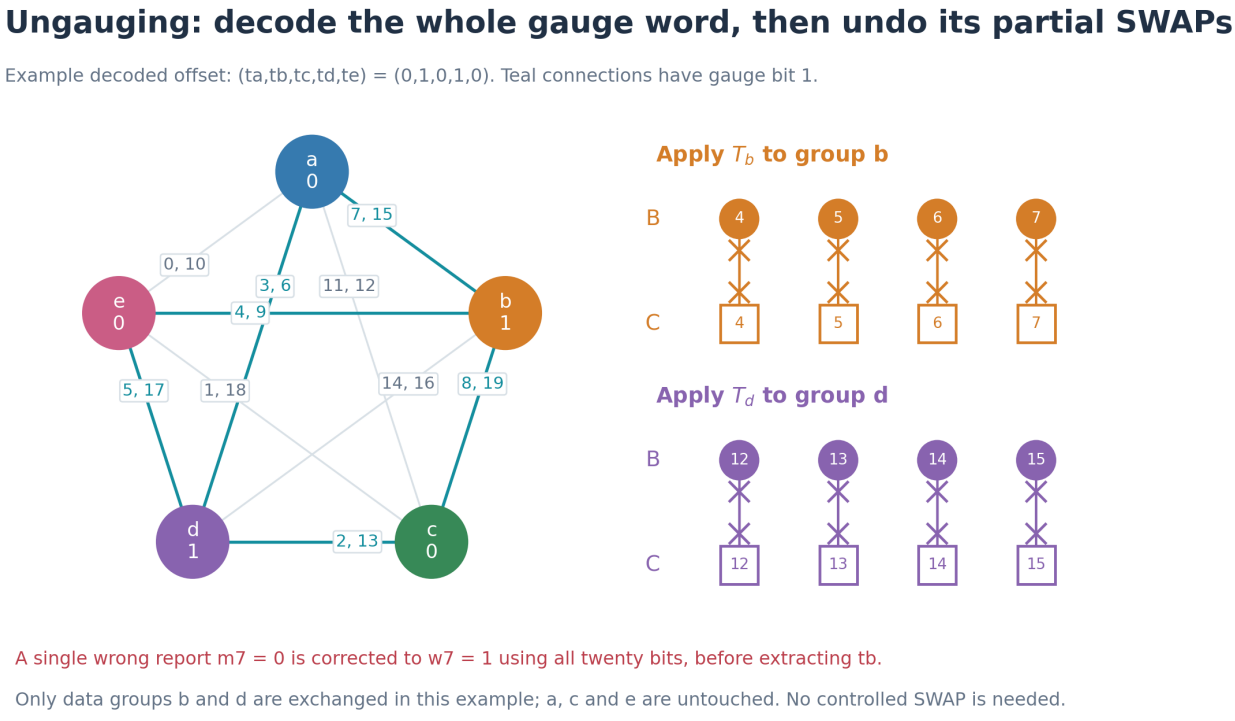


Figure 14: Worked decoding example: the corrected bits select b and d, with physical SWAPs on labels 4–7 and 12–15.

The five vertex labels in this drawing show the reconstructed classical bits. The physical vertices have already been measured; the drawing does not introduce five new quantum controllers. Its letter w denotes our decoded vector $\widehat{p}$, and its displayed vertex bits are our $\widehat{t}$.

12.5 Why the complete report has a unique correction

Proof steps 1 and 3 - protect the correction choice. The previous support bound limits the number of corrupt gauge bits. Decoding all twenty prevents those bits from selecting the wrong ordinary SWAPs.

Why the decoder is reliable. Before readout, each error term has at most one erroneous gauge bit per faulty location; there are no remaining vertex-to-gauge preparation CNOTs. Gauss star flips are changes between valid $p(t)$ strings. Each new readout fault changes at most its own reported position. Therefore

$$m = p(t) \oplus e, \qquad |e| \le 2$$

under the same global budget. For any choice of k vertices, there are $2k(5-k)$ edges joining them to the other vertices; the minimum nonzero count is eight. The difference of two valid strings has ones precisely on the edges separating the vertices whose t bits differ from those whose t bits agree. Distinct strings therefore differ on at least eight positions. Two candidates within distance two of m would differ on at most four positions, which is impossible.

Thus the decoder recovers the correct string for each term. Coherent terms sharing a report must share that decoded string. Gauge-Z phases become scalar amplitudes after readout. This establishes consistent feedforward without assuming stochastic, rather than coherent, errors.

13 Undo the partial data SWAPs

Ideal operation. Apply T_v when $\widehat{t}_v = 1$, using ordinary physical SWAPs, not coherent Fredkins. Visit physical labels q in order. Discard/reset A after its readout.

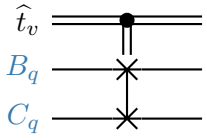


Figure 15: A decoded classical bit selects an ordinary SWAP of B_q and C_q.

For example, $\widehat{t} = 01000$ applies the four SWAPs on labels 4–7. On the corresponding component the ideal data operation is

$$V_t[V_t|\psi_+\rangle] = |\psi_+\rangle.$$

13.1 Following the two complementary vertex patterns

For the worked example in Section 12.4,

$$V_t = T_b T_d, \qquad V_{\bar{t}} = T_a T_c T_e = V_t U.$$

The same ancilla string $p(t)$ was present with both vertex components. Before the vertex-X measurements, their contributions after the enrichment Fredkins were

$$\begin{aligned} &|01010\rangle_V |p(t)\rangle_A \, T_b T_d |\psi\rangle \\ &+ |10101\rangle_V |p(t)\rangle_A \, T_a T_c T_e |\psi\rangle. \end{aligned}$$

These are two terms in one coherent state, not two observed computational-basis outcomes. Only these terms are displayed and their common normalization factor is omitted.

For all five vertex-X outcomes positive, $\langle+|^{\otimes 5}|01010\rangle = \langle+|^{\otimes 5}|10101\rangle = 1/\sqrt{32}$. After that readout, their common ancilla component carries the data sum

$$T_b T_d|\psi\rangle + T_a T_c T_e|\psi\rangle = V_t(I+U)|\psi\rangle.$$

Other accepted vertex-X outcomes give the same sum after the ancilla-Z corrections of Section 8. The ideal gauged checks do not change it. Later, measuring A selects its recorded component; full decoding recovers $p(t)$ and chooses $t_a = 0$ as above. The eight ordinary SWAPs selected by $V_t = T_b T_d$ act on physical pairs 4,5,6,7,12,13,14,15 and give

$$\begin{aligned} T_b T_d\big[T_b T_d|\psi\rangle + T_a T_c T_e|\psi\rangle\big] &= |\psi\rangle + U|\psi\rangle \\ &= 2\Pi_+|\psi\rangle. \end{aligned}$$

Had we chosen the complementary representative, its correction $V_{\bar{t}} = V_t U$ would instead give $U|\psi\rangle + |\psi\rangle$: the same state. Equivalently,

$$V_t\big[V_t|\psi_+\rangle\big] = |\psi_+\rangle, \qquad V_{\bar{t}}\big[V_t|\psi_+\rangle\big] = U|\psi_+\rangle = |\psi_+\rangle.$$

Thus $t_a = 0$ chooses a definite classical correction; it does not discard one of the two coherent contributions or infer a past vertex-Z outcome. The positive U eigenvalue established in proof step 2 is what makes the two ideal corrections equivalent.

13.2 Why the correcting SWAPs preserve the error bound

Section 12 determines which SWAPs to apply despite the allowed errors in the edge-measurement reports. We now check what those SWAPs do to physical errors already present on B and C, and what happens if a correcting SWAP itself fails.

First consider the example $t = (0,1,0,1,0)$**.** The required correction is $V_t = T_b T_d$. Without errors,

$$T_b T_d\big[T_b T_d|\psi_+\rangle\big] = |\psi_+\rangle.$$

Recall that T_b swaps $B_4, \ldots, B_7$ with $C_4, \ldots, C_7$, while T_d swaps positions 12–15.

Now put an X error on B_4 **before the correction.** The state entering this stage is

$$X_{B_4} T_b T_d|\psi_+\rangle.$$

Applying the correction gives

$$T_b T_d\big[X_{B_4} T_b T_d|\psi_+\rangle\big] = X_{C_4}|\psi_+\rangle.$$

The unwanted $T_b T_d$ has disappeared. The X error remains, but has moved from B_4 to C_4.

Both positions belong to **matched pair b**, consisting of the four-qubit block 4–7 in B and the corresponding block in C. The error has changed copies without reaching another matched pair.

The same reasoning applies to several errors. For example,

$$T_b T_d\big[X_{B_4} Z_{C_6} T_b T_d|\psi_+\rangle\big] = X_{C_4} Z_{B_6}|\psi_+\rangle.$$

Both errors remain in matched pair b.

The general calculation is the same. Let F denote a product of physical Pauli errors present just before the correction, so that the state is

$$FV_t|\psi_+\rangle.$$

After applying V_t, it becomes

$$V_tFV_t|\psi_+\rangle = F'|\psi_+\rangle, \qquad F' = V_tFV_t.$$

At each position q, conjugation by a correcting SWAP only exchanges the labels B and C:

$$X_{B_q} \leftrightarrow X_{C_q}, \qquad Y_{B_q} \leftrightarrow Y_{C_q}, \qquad Z_{B_q} \leftrightarrow Z_{C_q}.$$

The physical position q, and therefore its matched-pair label, stays unchanged. Thus an error confined to two matched pairs before the correction remains confined to those same two matched pairs afterwards.

Choosing the complementary vertex pattern does not change this bound. The alternative correction is $V_{\bar{t}} = UV_t$. On the ideal state, the additional complete SWAP U has no effect because

$$U|\psi_+\rangle = |\psi_+\rangle.$$

On an erroneous state it can exchange the errors between B and C. For example, the two correction choices applied to our erroneous input give

$$V_t[X_{B_4}V_t|\psi_+\rangle] = X_{C_4}|\psi_+\rangle,$$
$$V_{\bar{t}}[X_{B_4}V_t|\psi_+\rangle] = X_{B_4}|\psi_+\rangle.$$

These are generally different physical states, but both errors occupy matched pair b. The convention $t_a = 0$ simply fixes which correction we implement.

Finally, include faults in the correcting gates themselves. Each correcting gate is an ordinary SWAP between B_q and C_q. A fault in that gate may damage both targets, but both belong to one matched pair.

For example, suppose one earlier fault left damage in matched pair b. If a correcting SWAP between B_{13} and C_{13} then fails, it can add damage in matched pair d. The total affected set is now b and d.

These are the two faults allowed in the entire attempt. We do not allow two earlier faults followed by two more during correction.

Consequently, after this stage the unwanted partial SWAP has been removed, while every remaining error term is supported on at most two matched pairs. If a fault produces a superposition of Pauli errors, the calculation applies to each term separately. Different terms may affect different pairs, but each term satisfies the same two-pair bound.

This establishes **proof step 1 through the SWAP correction**. Section 14 then uses the original code checks and the two-group lemma to reject the damaging errors that remain.

14 Filter using the original B and C checks separately

Ideal operation. Measure the original 36-check family in three positive rounds, ordered B-X, B-Z, C-X, C-Z, increasing s in each:

$$\mathcal{O} = (S^X_{B,0:8}, S^Z_{B,0:8}, S^X_{C,0:8}, S^Z_{C,0:8}).$$

Use the ordinary Pauli gadgets below, extending target lists to the eight-body supports where required. All 108 check bits must be zero.

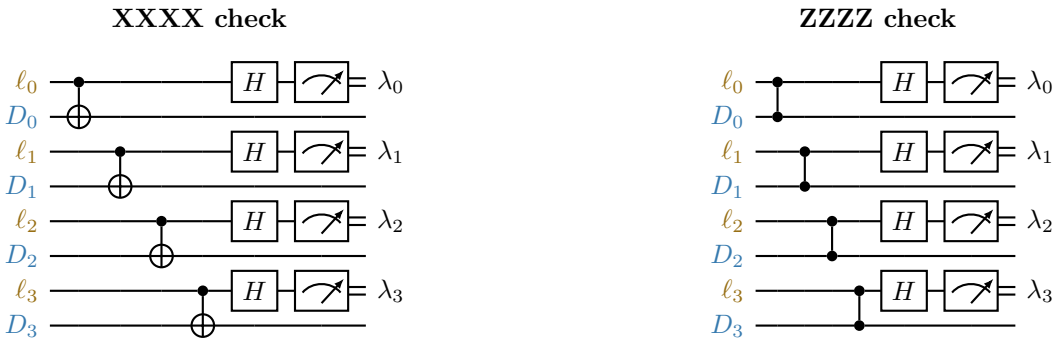


Figure 16: Original XXXX and ZZZZ circuits, now applied separately to each copy.

Since both copies of $|\psi_+\rangle$ are in their original code spaces, all these ideal measurements give $+1$ and leave the state unchanged.

The worked error is now detected. For $E = X_{B_0} X_{C_1} X_{C_2} X_{C_3}$,

$$S^Z_{B,0} E|\psi_+\rangle = -E|\psi_+\rangle, \qquad S^Z_{C,0} E|\psi_+\rangle = -E|\psi_+\rangle.$$

The paired product was positive, but both individual outcomes are negative. A clean original-code round rejects this component.

Proof step 3 - use the two-group lemma after restoring the original code. This is postselection, not recovery of an arbitrary unknown error on two entire inner blocks.

Why all larger inherited errors are harmless or rejected. For a Pauli term anticommuting with any original check S,

$$\frac{I+S}{2} E_\nu |\psi_+\rangle = 0.$$

For a term commuting with every original B/C check, apply the two-group lemma of Section 3 separately to each copy. It acts as a stabilizer up to phase, hence $E_\nu|\psi_+\rangle = \eta_\nu|\psi_+\rangle$. This is where the matched-pair support bound is used: physical weight alone could be as large as eight per copy, beyond the five-qubit detectability threshold supplied by distance six alone.

A complete clean round therefore maps the sum of error terms to

$$\sum_\nu c_\nu E_\nu |\psi_+\rangle \;\longmapsto\; c|\psi_+\rangle.$$

If $c = 0$, the branch is rejected; otherwise normalization gives the intended state. Harmless stabilizer errors are allowed to pass.

Among the three rounds, choose the last clean one as a proof reference. Every subsequent fault in these Pauli gadgets damages at most one physical data qubit, and

$$(aI + bS)F|\psi_+\rangle = (aF + bSFS)|\psi_+\rangle$$

does not enlarge F's support. All SWAP gates have finished. Thus at the end of this stage only small errors caused after that clean round remain, within the remaining fault budget. A late X_{C_0}, after the last interaction able to detect it, can escape the filter with all reports positive. Recovery is still needed.

15 Recover with repeated syndromes and single-qubit corrections

Ideal operation. Measure the same joint 36-bit family $\mathcal{O}$ using fresh verified cats. Retain either check sign, apply no correction between rounds, and stop at the first three consecutive identical records. Allow at most nine rounds. Reject failed cat verification, failure to stop, or absence of a weight-at-most-two correction.

On each copy choose the fixed minimum-weight Pauli of weight at most two matching its eighteen stopping bits. Apply its single-qubit factors in physical-label order, B then C; omit identity factors. In the noiseless case the first three records are zero and the correction is I.

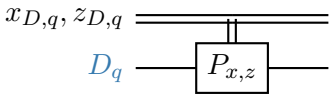


Figure 17: The classical decoder selects a single-qubit Pauli at each data position.

In this diagram $P_{0,0} = I,\ P_{1,0} = X,\ P_{0,1} = Z,\ P_{1,1} = Y$. For Y, use one physical Y gate. The lookup is classical, not a coherent many-qubit operation.

Example. A late X_{C_0} produces negative C Z checks at s=0 and s=5, and no negative X checks; all B checks are positive. A stable record of that syndrome selects $R_C = X_{C_0}$, which cancels it.

Proof step 4 - choose a logically correct recovery despite noisy records. The earlier filter has reduced the inherited error class from whole matched pairs to at most two physical qubits. Only this smaller class is passed to the recovery lookup.

Why the reports suffice. Of a stopping triple, at least one round is clean. Its common record is a true syndrome at that reference time. Nine rounds contain three consecutive clean ones: with two bad rounds, at most three clean runs of length two would give only eight rounds. No decoder is told which rounds are clean.

At the reference time, each copy's Pauli error F has weight at most two. The selected correction R has the same syndrome, so RF commutes with every check and has weight at most four. Distance six makes it a stabilizer up to phase:

$$RF|\psi_+\rangle = e^{i\theta}|\psi_+\rangle.$$

Later errors remain local through Pauli corrections. For example, if a new X fault on C_0 occurs after its detecting interactions in the third round, three zero records may still select I. The residual X_{C_0} is not magically removed, but is a correctable physical error. Correction-gate faults and later waiting faults are counted in the remaining global allowance.

16 Read B, decode the reports, and retain magic in C

Ideal operation. Measure all twenty B qubits in Z. Define the nine parities of a binary report x by

$$h_s(x) = \bigoplus_{q \in R_s} x_q, \qquad s = 0, \ldots, 8.$$

For the full reported word n, choose a minimum-weight bit-flip mask e with $|e| \le 2$ and $h(e) = h(n)$. Reject if no candidate exists. Break ties lexicographically by flipped-position lists. Set $n' = n \oplus e$ and keep only

$$\ell = n'_0 \oplus n'_3 \oplus n'_6 \oplus n'_7 \oplus n'_{18} \oplus n'_{19} = 0.$$

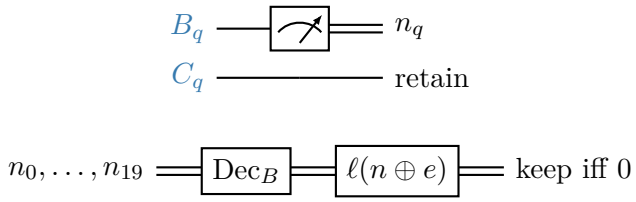


Figure 18: Physical B-Z readout and classical decoding; C remains a quantum output.

On the first logical pair, with both second logical qubits in zero,

$$\frac{(I+U)|0+\rangle}{\|(I+U)|0+\rangle\|} = \frac{2|00\rangle + |01\rangle + |10\rangle}{\sqrt{6}}.$$

Selecting decoded logical B zero gives

$$\frac{2|0\rangle + |1\rangle}{\sqrt{5}} \quad \text{in logical } C_0.$$

Discard B. The twenty-qubit C block is the retained resource.

Readout fault example. An allowed ideal physical report is all zeros. If only its position zero is misreported, the raw six-position logical parity is wrong. Full decoding finds the single-position mask e at zero, returns $n' = 0^{20}$, and gives the correct logical outcome. An X or Y error before readout acts like another flipped bit. A Z-only error on measured B contributes only a scalar in that branch:

$$\langle n|_B Z_{B_q}|\Psi\rangle = (-1)^{n_q}\langle n|_B|\Psi\rangle.$$

Thus it does not alter the retained C state for that physical result.

Proof step 4 - include readout faults and C waiting errors. No error-free measurement or final recovery is assumed after this point.

Why decoding preserves the logical decision. The combined B bit errors and wrong reports occupy at most two positions under the same remaining budget. Two masks of weight at most two with the same Z-check syndrome differ by an X Pauli of weight at most four commuting with all Z checks and automatically all X checks. Distance six makes it a stabilizer, which cannot change the logical-Z parity. The chosen and true corrections therefore give the same logical decision.

C waiting errors during B readout are included. For example, an X fault on C_0 can leave $X_{C_0}|\overline{M0}\rangle$. It is a physical error in the correctable space, not a wrong logical resource. The claim is correctness of the decoded C state, without adding an unbudgeted fault-free physical recovery after the circuit.

17 What the complete argument establishes

Known-state verification removes arbitrary raw-source errors or rejects. The subsequent circuits charge at most one new matched-pair label per fault. A clean gauged round fixes the positive logical SWAP projection. Full gauge decoding makes the SWAP correction consistent and preserves the local-error labels. A clean original-code filtering round removes the inherited two-matched-pair damage or rejects. All later errors are physical-qubit errors handled by noisy recovery and decoded readout within the same two-fault budget.

Therefore every accepted attempt with at most two faulty locations has decoded C state $|M\rangle \otimes |0\rangle$. The number of faults, the number of damaged physical qubits, and the number of affected inner blocks are distinct quantities throughout this argument.

With serial helper reuse, 77 physical qubits suffice: 40 data, 20 gauge, and 17 helpers. Five helpers initially serve as vertices. Resource checks need at most nine leaves plus w; after vertex readout the largest paired check needs sixteen leaves plus w. The Fredkin count is

$$20 + 3(5 \cdot 4) = 80.$$

Ungauging adds at most sixteen ordinary SWAPs. The syndrome recovery uses at most nine rounds; all its faults and idles are included. In the absence of faults, conditioned on successful input preparation, positive SWAP selection succeeds with probability 3/4, and logical extraction with conditional probability 5/6, giving 5/8 overall. This is not a noisy-yield or optimal-overhead claim.

A Physical supports and logical operators

The following rows specify both all-X and all-Z original checks. Measure only s=0 through 8 as independent code checks. Row 9 is their redundant upper row used in the graph construction.

$$
\begin{aligned}
R_0 &= \{0,1,2,3\}, & R_1 &= \{4,5,6,7\},\\
R_2 &= \{8,9,10,11\}, & R_3 &= \{12,13,14,15\},\\
R_4 &= \{16,17,18,19\},\\
R_5 &= \{0,3,6,7,10,11,12,15\},\\
R_6 &= \{4,7,8,9,14,15,16,19\},\\
R_7 &= \{1,2,8,11,12,13,18,19\},\\
R_8 &= \{2,3,5,6,13,14,16,17\},\\
R_9 &= \{0,1,4,5,9,10,17,18\}.
\end{aligned}
$$

The stars are $E_a = R_5, E_b = R_6, E_c = R_7, E_d = R_8, E_e = R_9$. For either copy D, the logical operators needed are

$$
\begin{aligned}
\overline{X}_0^D &= \prod_{q\in\{2,3,8,11,12,15\}} X_{D_q},\\
\overline{Z}_0^D &= \prod_{q\in\{0,3,6,7,18,19\}} Z_{D_q},\\
\overline{Z}_1^D &= \prod_{q\in\{2,3,8,11,12,15\}} Z_{D_q}.
\end{aligned}
$$

B Sources and reproducibility

The manuscript's Appendix A gives the C4-Helix construction and Table-I stabilizers. The original SWAP-gauging construction and Magic Quantum Code Surgery supply the background. The specific postselected single-cycle schedule and its two-fault argument are the construction developed in these notes; they are not obtained by directly quoting a general distance-six circuit theorem from those papers.

The standalone proof, equation-level procedure, and circuit atlas remain available. The classical acceptance and correction routines are implemented in the decoder. The finite certificate script checks code, graph, decoding, and algebraic ingredients, not every physical fault pair. The integrated-note builder reuses the atlas's original quantikz circuits and builds the LaTeX, PDF, and circuit images for this note.